SUPERSEDING
MIL-W-12995A(EL)
1 July 1965

MILITARY SPECIFICATION

WIRE, ELECTRICAL: ANTENNA
(WIRE, W-120)

1. SCOPE

1.1 This specification covers stranded, round electrical wire fabricated from silicon bronze, suitable for antenna applications.

1.2 Classification. - Silicon bronze wire covered by this specification shall be of the following type: (See 6.2.)

1.2.1 Wire W-120 consists of 42 strands of No. 32 AWG (0.008") silicon bronze wire, twisted in 6 groups of 7 each, and then twisted around a three-ply or five-ply cotton thread core forming one complete cylindrical conductor.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-12000 Cable, Cord, and Wire, Electric; Packaging of.

STANDARDS

MILITARY

MIL-STD-105 Sampling Inspection Tables for Inspection by Attributes.

(Copies of specifications, standards, and other publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

FSC 6145

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3. REQUIREMENTS

3.1 Material. - The material for the wire shall be as specified herein. The best material commercially available for the purpose shall be used when a definite material is not designated.

3.2 Strands. - Each strand shall consist of a silicon bronze alloy wire having a diameter of 0.008 inch ± 0.001 inch (No. 32 A.W.G.).

3.3 Conductor construction. - The stranded wire conductor shall be constructed of 42 strands of wire as specified in 3.2 The strands shall first be twisted into 6 groups of seven strands each, then the 6 groups shall be twisted around a three-ply or five-ply cotton thread core so as to form one complete cylindrical stranded conductor.

3.3.1 Joints in conductors. - Factory joints in conductors shall be avoided, and in no case shall they exceed two joints per strand per length of wire, with joints spaced not less than 3 feet apart. All joints shall be butt brazed fusing with silver alloy solder and a nonacid flux, and shall be finished smoothly in a workmanlike manner. No twist joints shall be used. The tensile strength of a section of each strand, which includes the joint, shall be not less than 90 percent of the tensile strength of an adjacent section of the strand of equal length without the joints.

3.3.2 Tensile strength. - The tensile strength of the silicon bronze stranded conductor shall be not less than 200 pounds.

3.3.3 Elongation. - The elongation shall be not less than 0.80 percent for a 60 inch length of stranded conductor.

3.3.4 Continuity. - Each conductor shall indicate continuity when tested as specified in 4.7.

3.4 Workmanship. - Strands and completed wire shall be manufactured and assembled in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Operator qualification test. (See 4.3.)
(b) Quality conformance inspection before preparation for delivery. (See 4.4.)
(c) Quality conformance inspection of preparation for delivery. (See 4.8.)

4.3 Operator qualification test. - This test shall be performed at the start of production for each order for wire under this specification and shall be as follows: Each operator required to make joints in strands will submit a group of twelve (12) specimens of strand which include a joint with a specimen of adjacent normal section of each. These specimens will be tested for breaking strength. An operator will be considered qualified after submitting a group of twelve specimens which meet the requirements of 3.3.1. Records of such qualifications shall be available for verification by the Government.

4.4 Quality conformance inspection before preparation for delivery. The supplier shall perform the inspection specified in 4.4.1 and 4.4.2. This does not relieve the supplier of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the supplier's inspection procedures and examine the supplier's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the supplier's compliance with specification requirements. Test equipment for Government verification inspection shall be made available by the supplier.

4.4.1 Group A inspection. - Group A inspection shall consist of the examinations and tests specified in table I. Statistical sampling shall be in accordance with the ordinary inspection procedure of Standard MIL-STD-105, AQL-1.0% for major defects and 4.0% for minor defects.

4.4.2 Group B inspection. - Group B inspection shall consist of the examinations and tests specified in table I. They shall be performed on specimens from sample units that have been subjected to and have passed the Group A tests, unless the Government considers it more practical to select a sample from the inspection lot for Group B tests. Sampling shall conform to the special level S-4 for normal, tightened, and reduced inspection of MIL-STD-105. AQL shall be 4.0%.

Table I - Government inspection

	Requirement paragraph	Test paragraph	Group A AQL		Group B AQL
			Major	Minor	
Visual and dimensional inspection	3.2 and 3.3	4.5	1%	4%	
Continuity	3.3.4	4.7	1%	-	
Tensile strength	3.3.2	4.6			4%
Elongation	3.3.3	4.6			4%

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4.5 Visual and dimensional inspection. - Strands, and completed wire shall be inspected to verify that the materials and all workmanship comply with this specification. At the option of the Government, certification as to materials and splices may be accepted. (See 3.2 and 3.3.)

4.5.1 Dimensional inspection.

4.5.1.1 Strand diameter. - Strands shall be measured with a micrometer at three places, one near each end and one in the middle of the wire. The average of the measurements obtained on any piece shall be used for the complete unit.

4.5.1.2 Construction. - Construction shall be determined by physical examination and measurement of the complete wire.

4.6 Tensile-strength and elongation. - Tensile-strength and elongation shall be made on a tensile-strength machine, preferably of the pendulum types. The tensile strength test shall be made after stranding, on the completed wire. The free length between grips of the test specimen shall be 60 inches $\pm$ 1 inch, and the speed shall be not more than 3 inches per minute. Care shall be taken to insure that strands in the stranded wire are evenly gripped during test. (Successful testing of stranded wire for tensile properties requires an adequate means of gripping the ends of the test specimen. Various means are available, such as a long tube or socket into which the wire may be soldered, or in which, after insertion, the wires may be swaged or pressed without serious distortion. Ordinary jaws or clamping devices are usually not suitable.) The breaking strength of strands shall be computed by using the nominal diameters and tensile strength specified in 3.2 and 3.3.2. Elongation tests shall be made on strands taken as specified in 4.5.1 and shall meet the requirements of 3.3.3.

4.7 Continuity. - Each complete length of Wire W-120 shall be tested for continuity using a potential of not more than 10 volts d.c.

4.8 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Wires shall be preserved, packaged, packed and marked in accordance with Specification MIL-C-12000. (See 6.2.)

5.2 Preservation, packaging and packing shall be level A, B, or C as specified. (See 6.2.)

6. NOTES

6.1 Intended use. - The wire covered by this specification is most commonly used in antenna applications.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of wire and length required. (See 1.2.)
(c) Size of spool required and length on each when applicable.
(d) Level of packaging and level of packing required. (See 5.1, 5.2.)

6.3 Definitions.

6.3.1 Inspection lot. - An inspection lot shall consist of all units of product produced under essentially the same conditions and submitted for acceptance at the same time.

6.3.2 Unit of product. - A unit of product is the continuous length of wire contained on a single reel, or spool, or in a package.

6.3.3 Specimen. - A specimen consists of the individual piece of the unit of product upon which the test is made.

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Project No. 6145-A010

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Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-W-12995B(EL) WIRE, ELECTRICAL: ANTENNA (WIRE, W-120)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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